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Essential Unix Commands

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Unix commands are instructions used to interact with the Unix operating system through a terminal, allowing users to manage files, processes, users, and system resources efficiently. Unix is a powerful, multi-user, and multi-tasking operating system originally developed at Bell Labs and later inspired many modern systems, including Linux.

- Unix was developed in the 1970s at AT&T Bell Labs by Dennis M. Ritchie and Ken Thompson.
- The C programming language was created specifically to build the Unix operating system.
- Linux is a Unix-like operating system that follows the same design philosophy and command

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- The core of Linux is the Linux Kernel, first released in the early 1990s by Linus Torvalds.
- All Unix/Linux commands are case-sensitive, meaning Hello and hello are treated differently.

Below are several sections that explain the most essential Linux commands:

1. File System Navigation Unix Commands

Command	Description	Example
cd	Changes the current working directory.	cd Documents
ls	Lists files and directories in the current directory.	ls
pwd	Prints the current working directory.	pwd
mkdir	Creates a new directory.	mkdir new_folder
rmdir	Removes an empty directory.	rmdir empty_folder
mv	Moves files or directories.	mv file1.txt Documents/

2. File Manipulation Unix Commands

Command	Description	Example
touch	Creates an empty file or updates the access and modification times.	touch new_file.txt
cp	Copies files or directories.	cp file1.txt file2.txt

Command	Description	Example
mv	Moves files or directories.	mv file1.txt Documents
rm	Remove files or directories.	rm old_file.txt
chmod	Changes the permissions of a file or directory.	chmod 644 file.txt
chown	Changes the owner and group of a file or directory.	chown user:group file.txt
ln	Creates links between files.	ln -s target_file symlink
cat	Concatenates files and displays their contents.	cat file1.txt file2.txt
head	Displays the first few lines of a file.	head file.txt
tail	Displays the last few lines of a file.	tail file.txt
more	Displays the contents of a file page by page.	more file.txt
less	Displays the contents of a file with advanced navigation features.	less file.txt
diff	Compares files line by line.	diff file1.txt file2.txt
patch	Applies a diff file to update a target file.	patch file.txt < changes.diff

3. Process Management Unix Commands

Command	Description	Example
ps	Displays information about active processes, including their status and IDs.	ps aux
top	Displays a dynamic real-time view of system processes and their resource usage.	top
kill	Terminates processes using their process IDs (PIDs).	kill <pid>
pkill	Sends signals to processes based on name or other attributes.	pkill -9 firefox
killall	Terminates processes by name.	killall -9 firefox
renice	Changes the priority of running processes.	renice -n 10 <pid>
nice	Runs a command with modified scheduling priority.	nice -n 10 command

Command	Description	Example
pstree	Displays running processes as a tree.	pstree
pgrep	Searches for processes by name or other attributes.	pgrep firefox
jobs	Lists active jobs and their status in the current shell session.	jobs
bg	Puts a job in the background.	bg <job_id>
fg	Brings a background job to the foreground.	fg <job_id>
nohup	Runs a command immune to hangups, with output to a specified file.	nohup command &
disown	Removes jobs from the shell's job table, allowing them to run independently.	disown <job_id>

4. Text Processing Unix Commands

Command	Description	Example
grep	Searches for patterns in text files.	grep "error" logfile.txt
sed	Processes and transforms text streams.	sed 's/old_string/new_string/g' file.txt
awk	Processes and analyzes text files using a pattern scanning and processing language.	awk '{print $1, $3}' data.csv

5. Network Communication Unix Commands

Command	Description	Example
ping	Tests connectivity with another host using ICMP echo requests.	ping google.com
traceroute	Traces the route that packets take to reach a destination.	traceroute google.com
nslookup	Queries DNS servers for domain name resolution and IP address information.	nslookup google.com
dig	Performs DNS queries, providing detailed information about DNS records.	dig google.com
host	Performs DNS lookups, displaying domain name to IP address resolution.	host google.com

Command	Description	Example
whois	Retrieves information about domain registration and ownership.	whois google.com
ssh	Provides secure remote access to a system.	ssh username@hostname
scp	Securely copies files between hosts over a network.	scp file.txt username@hostname:/path/
ftp	Transfers files between hosts using the File Transfer Protocol (FTP).	ftp hostname
telnet	Establishes interactive text-based communication with a remote host.	telnet hostname
netstat	Displays network connections, routing tables, interface statistics, masquerade connections, and multicast memberships.	netstat -tuln
ifconfig	Displays or configures network interfaces and their settings.	ifconfig
iwconfig	Configures wireless network interfaces.	iwconfig wlan0
route	Displays or modifies the IP routing table.	route -n
arp	Displays or modifies the Address Resolution Protocol (ARP) cache.	arp -a
ss	Displays socket statistics.	ss -tuln
hostname	Displays or sets the system's hostname.	hostname
mtr	Combines the functionality of ping and traceroute, providing detailed network diagnostic information.	mtr google.com

6. Coding and Development Unix Commands

Command	Description	Example
vim / nano	Used to write and edit source code files from the terminal.	vim main.c
gcc	Compiles C programs into executable files.	gcc program.c -o program
make	Automates compilation using a Makefile.	make
./program	Runs a compiled executable file.	./program
python3	Executes Python scripts.	python3 script.py

Command	Description	Example
git	Manages source code version control.	git status
grep	Searches code files for specific patterns or keywords.	grep "main" *.c

7. System Administration Unix Commands

Command	Description	Example
df	Displays disk space usage.	df -h
du	Displays disk usage of files and directories.	du -sh /path/to/directory
crontab -e	Manages cron jobs, which are scheduled tasks that run at predefined times or intervals.	crontab -e

8. Text Editors in Unix

Text Editor	Description	Example
Vi / Vim	Vi (Vim) is a highly configurable, powerful, and feature-rich text editor based on the original Vi editor. Vim offers modes for both command-line operations and text editing.	Open a file with Vim: vim filename Exit Vim editor: Press Esc, then type :wq and press Enter
Emacs	Emacs is a versatile text editor with extensive customization capabilities and support for various programming languages.	Open a file with Emacs: emacs filename Save and exit Emacs: Press Ctrl + X, then Ctrl + S and Ctrl + X, then Ctrl + C to exit
Nano	Nano is a simple and user-friendly text editor designed for ease of use and accessibility.	Open a file with Nano: nano filename Save and exit Nano: Press Ctrl + O, then Ctrl + X
Ed	Ed is a standard Unix text editor that operates in line-oriented mode, making it suitable for batch processing and automation tasks.	Open a file with Ed: ed filename Exit Ed editor: Type q and press Enter
Jed	Jed is a lightweight yet powerful text editor that provides an intuitive interface and support for various programming languages.	Open a file with Jed: jed filename Save and exit Jed: Press Alt + X, then type exit and press Enter

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